Hyperspectral Unmixing Using Higher-Order Graph Regularized NMF With Adaptive Feature Selection

Kewen Qu, *Member, IEEE*, Zhenqing Li, *Student Member, IEEE*, Chenyang Wang, Fangzhou Luo, and Wenxing Bao, *Member, IEEE*

***Abstract*— Recently, graph learning methods have attracted much research attention, which uses first-order nearest-neighbor relation between pixels to construct adjacency graphs for capturing smooth abundance. However, the first-order nearest-neighbor information neglects the shared domain structure of pixels making the higher-order nearest-neighbor relation missing resulting in limited spatial structure learning. Additionally, the feature values in hyperspectral images (HSIs) vary greatly in different bands, tiny value contributions are easily neglected due to the statistical properties of the model, which hinders the learning efficiency of the approach. To address these shortcomings, a higher-order graph regularizer nonnegative matrix factorization (NMF) with adaptive feature selection method is proposed. Specifically, we introduce pixel second-order nearest-neighbor relation in graph learning to capture the nearest-neighbor domains of pixels with missing connection in first-order neighborhoods and enhance the aggregation of homogeneous regions. Then, the first-order and second-order nearest-neighbor relation are combined to construct higher-order graph regularizer to better preserve the global spatial structure of the image. Additionally, adaptive weight is introduced in the data reconstruction to balance the effects of different feature values by adaptively selecting spectral features to enhance the contribution of tiny features. Finally, the objective function is designed by fusing the higher-order graph regularization, adaptive feature selection, and $L_{1/2}$ sparse regularizer into the NMF framework, and the optimization algorithm of the model is derived using a multiplicative update rule (MUR). We conducted extensive experiments on synthetic and real datasets, and the effectiveness and superiority of the proposed method were verified by comparing it with several state-of-the-art methods.**

***Index Terms*— Feature selection, higher-order graph learning, hyperspectral unmixing (HU), nonnegative matrix factorization (NMF).**

I. Introduction

With the development of hyperspectral imaging technology, hyperspectral sensors acquire more and more detailed information in the spectral band, which provides a strong guarantee for ground observation. Hyperspectral images (HSIs) are widely used in the fields of agricultural yield estimation, mineral exploration, and environmental monitoring [1]. Due to the limitation of low spatial resolution of sensors, different substances inevitably mix together to form "mixed pixels," which severely limits the application of HSIs in fine classification [2], target detection [3], and other fields. Therefore, the mixed pixels hinder the exploration of fine material discrimination, which makes hyperspectral unmixing (HU) an important step in HSIs, whose main purpose is to identify pure objects (i.e., endmembers) and their proportion (i.e., abundances) from the mixed pixels.

For HU issues, a mass of approaches have been proposed based on linear mixing model (LMM), which can be roughly divided into geometrical, statistical, sparse regression, and deep learning (DL) based methods. Specifically, the geometrical-based approaches are dependent on the particular geometric properties of the hyperspectral data in the feature space, which mainly includes the subspace projection methods [4], the maximum simplex [5], and the minimum simplex methods [6]. The sparse regression-based approach is an accurate estimation of abundance with a library of pure endmember spectra. Iordache et al. [7] designed a fast abundance estimation algorithm (SUnSAL) using the spectral information contained in the original data and applying L_1-norm constraints to the abundance matrix. Subsequently, in order for the model abundance coefficients to maintain the spatial relation of the original data, Iordache et al. [8] designed an abundance estimation method based on joint spatial–spectral information to improve the unmixing performance of the sparse regression model. The DL approaches can obtain the hierarchical structure in hyperspectral data, which overcomes the limitations of single-layer information. Li et al. [9] proposed a self-supervised robust deep matrix factorization unmixing method. Feng et al. [10] proposed sparsity-constrained deep nonnegative matrix factorization (L_1-DNMF), which the unmixing process is divided into two steps, preprocessing and fine-tuning, to better learned the decomposed matrices and improve the accuracy of abundance estimation.

In statistical frameworks, nonnegative matrix factorization (NMF) has become a representative and powerful method due to its ability to estimate both endmembers and abundances, and to efficiently handle highly mixed data. NMF decomposes high-dimensional data into the product of two nonnegative low-rank matrices, called "basis" and

Manuscript received 15 February 2023; revised 24 March 2023 and 8 April 2023; accepted 4 May 2023. Date of publication 12 May 2023; date of current version 25 May 2023. This work was supported in part by the Natural Science Foundation of Ningxia Province of China under Project 2021AAC03179 and Project 2020AAC02028; and in part by the Science Foundation of North Minzu University, Yingchuan, China, under Project 2020KYQD50. *(Corresponding author: Kewen Qu.)*

The authors are with the Image and Intelligence Information Processing Innovation Team, National Ethnic Affairs Commission, Yinchuan 750021, China, and also with the School of Computer Science and Engineering, North Minzu University, Yinchuan 750021, China (e-mail: kewen.qu@nmu.edu.cn; li-zhenqing@outlook.com; w870352094@163.com; luofangzhou126@163.com; bwx71@163.com).

Digital Object Identifier 10.1109/TGRS.2023.3275740

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

"representation coefficient" [11], which perfectly fit the spectral mixing mechanism and have been widely studied. Unfortunately, due to the nonconvexity of the model, the solution space is huge and there are a large number of local solutions, which is not suitable for direct application to the unmixing problem.

For increasing the unmixing accuracy, in the past decade or so, numerous priors such as abundance sparsity and smoothness, spatial structure, and manifold learning have been introduced into the unmixing models. Based on the sparse prior, Qian et al. [12] proposed $L_{1/2}$ sparsity-constrained NMF to enhance the abundance sparsity by $L_{1/2}$ regularizer and proved that $L_{1/2}$ regularizer is easier to solve than L_0 regularizer and have stronger sparsity than L_1 regularizer, and is a better alternative choice for $L_p(0 < p < 1)$ regularizer. Miao and Qi [13] first introduced the minimum volume regularization into the NMF framework, by finding the simplex that can enclose all pixel points and has the smallest volume for HU. He et al. [14] proposed a reweighted sparse NMF with total variation (TV) regularized, and assumed that the value of the weight is inversely proportional to the value of the abundance. Qu and Bao [15] proposed a multiple-priors ensemble constrained NMF unmixing method that integrated the minimum volume constraint (MVC), reweighted $L_{1/2}$-norm and TV regularization into the NMF framework, and imposing constraints on the endmember and the abundance matrix in the NMF model, which overcomes the limitations of the traditional single matrix constraint method.

In recent years, with the rise of data manifold research [16], [17], [18], hyperspectral data are more likely to lie on low-dimensional submanifold embedded in the high-dimensional data space. Therefore, the intrinsic manifold structure of HSIs should be considered when unmixing, and the corresponding embedding mapping should be obtained by recovering the low-dimensional intrinsic submanifold from the HSI data, so as to maintain the close link between the original image and the material abundance maps. Lu et al. [19] proposed sparse NMF with graph regularized inspired by manifold learning and sparse algorithm, which captures the inherent local geometric structure of HSIs through constructing the nearest-neighbor graph of pixels, so as to smooth the abundance and suppress noise interference. Rathnayake et al. [20] introduced a novel Laplacian regularization based on the L_1-norm, which is proven to be superior to the L_2-norm in piecewise smooth and noise rejection capabilities by graph spectral analysis. Similarly, in [21], the latent characteristic structure is captured by subspace clustering, for which the weight matrix is constructed by utilizing the self-expression coefficients of the image itself, which can circumvent the effect of noise when constructing the similarity graph. Zhu et al. [22] proposed a structured sparse unmixing method that uses spectral angular distance similarity to obtain the nearest feature distance to avoid images spanning different pixels, thus preserving edge information.

However, the above graph learning-based approach only considers the use of first-order nearest-neighbor relation (i.e., the similarity matrix is constructed by only considering the first k-nearest pixels) to obtain smooth abundance. The first-order nearest-neighbor information observed in the real HSIs is not sufficient to maintain the global spatial structure and has not fully mined the neighborhood relation between pixel points [23], i.e., two pixel points are essentially similar but have no first-order nearest-neighbor information, such a situation we call missing higher-order nearest-neighbor relation. In this article, the second-order nearest-neighbor relation between pixel points is investigated, which is not determined by the magnitude of the connection weight, but is constructed from the shared neighborhood structure of the pixels. The second-order nearest-neighbor relation can be explained by the fact that pixel points with shared neighborhoods are likely to be similar. Second-order nearest-neighbor relation is considered in graph learning to complement pixels with missing connection in first-order nearest-neighbor relation, to enhance the aggregation of homogeneous regions, and to fully maintain the global spatial structure of HSIs [24].

In addition, the spectral dimension as an important feature in HSIs to distinguish between different endmembers. The reflectance is often unequal on different bands per pixel. These tiny features are usually in bands with low reflectivity and they have a high chance to be suppressed or even neglected in the model reconstruction due to the interference of outliers and noise. Therefore, it is not optimal to use the original hyperspectral data directly in HSIs reconstruction [25]. To this end, we propose an NMF reconstruction model with adaptive feature selection. The model generates an improved spectral feature space by introducing an adaptive weight matrix to select appropriate weight for different bands of each pixel, enhancing the contribution of minute features to the statistical performance of the NMF to improve the discrimination of endmembers.

In this article, second-order nearest-neighbor relation is introduced to enhance the aggregation of homogeneous regions, in order to capture the nearest-neighbor relation of pixels with missing connections in the first-order nearest-neighbor relation information, and constructing higher-order Laplacian graph regularization term by combining first-order and second-order nearest-neighbor information to better preserve the global spatial structure. In order to balance the influence of different feature values and to achieve adaptive spectral feature selection, we introduced the adaptive weight matrix into the NMF model to enhance the discriminative power of the endmembers. Based on the above analysis, we propose a higher-order graph regularization NMF with adaptive feature selection (HGNMF-FS) for HU.

The main contributions of this article are summarized as follows.

1) We propose a new blind approach for HU, i.e., an HGNMF-FS. The model incorporates higher-order graph regularizer, adaptive feature selection weights, and $L_{1/2}$ sparse regularizer, and achieves improved unmixing performance by mining spatially shared information in the pixel neighborhood, the contribution of tiny spectral feature values, and joint abundance sparsity constraints, respectively.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

2) Second-order nearest-neighbor relation of pixel neighborhood and higher-order graph Laplacian regularizer are introduced in HU. By capturing the missing connections in the first-order nearest-neighbor relation, it fills the shortage of first-order graph learning and better preserves the global spatial structure of HSIs by jointly using the first-order and second-order nearest-neighbor graph information.
3) Introducing adaptive reweighting in the spectral dimension of HSIs improves the model's attention to tiny features, and mitigates the adverse effects of different feature values on data reconstruction by performing unmixing in the corrected spectral subspace.

The remainder of this article is organized as follows. Section II briefly explains LMM and the basic NMF model. Sections III and IV describe the model of the proposed algorithm and the corresponding optimization algorithm. Section V describes and analyzes the results on synthetic datasets and real datasets. In Section VI, the conclusion is brought.

II. Background

In this section, we will briefly introduce NMF method for HU, which includes LMM and sparse NMF. In the following summary, matrices and vectors are represented by bold capital letters and lowercase letters, respectively. For convenience, we use capital $\boldsymbol{X} = [x_1, x_2, x_3, \ldots, x_N] \in \mathbb{R}^{L\times N}$ to represent HSIs, with L bands and N pixels.

A. Linear Mixing Model

LMM is an essential framework to deal with HU, and it assumes that spectra of distinct materials are not interfered with each other [26], [27]. Therefore, a mixed pixel can be approximated by decomposing two matrices, one being base matrix, called the endmember, and the other a coefficient matrix, called the abundance. The LMM is defined as

$$x_i = \boldsymbol{A}s_i + e_i, \quad (i = 1, 2, \ldots, N) \tag{1}$$

where x_i denotes the ith pixel in the HSIs. $\boldsymbol{A} = [a_1, a_2, \ldots, a_M] \in \mathbb{R}^{L\times M}$ denotes the endmember matrix; $s_i \in \mathbb{R}^{M\times 1}$ denotes the abundance vector of pixel x_i, $e_i \in \mathbb{R}^{L\times 1}$ represents the residual term. The number of endmember M should be noted to be much smaller than the number of bands in the HSIs, i.e., $M \ll L$.

With respect to the HSIs, LMM can be further expressed as

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S} + \boldsymbol{E} \tag{2}$$

where $\boldsymbol{S} \in \mathbb{R}^{M\times N}$ denotes abundance matrix, and $\boldsymbol{E} \in \mathbb{R}^{L\times N}$ denotes the residual matrix. In LMM, abundance matrix should satisfy the abundance sum-to-one constraint (ASC) and abundance nonnegative constraint (ANC), respectively. Nonnegative constraint is to ensure that each endmember and abundance conform to the physical meaning, while sum-to-one is to indicate that the sum of abundances of each endmember in pixel should be one. Specifically, they can be expressed by the formula as follows:

$$\begin{aligned} &\text{ANC}: s_{ij} \geq 0, \quad (i = 1, 2, \ldots, M, j = 1, 2, \ldots, N) \\ &\text{ASC}: \sum_{i=1}^{M} s_{ij} = 1, \quad (j = 1, 2, \ldots, N). \end{aligned} \tag{3}$$

B. Sparse NMF

NMF is a widely used method in the field of data analysis, especially when nonnegative constraint is considered. The purpose of NMF is to decompose the nonnegative HSIs matrix $\boldsymbol{X} \in \mathbb{R}^{L\times N}$ into the product of two nonnegative low-rank matrices, and NMF can be written as

$$\boldsymbol{X} \approx \boldsymbol{A}\boldsymbol{S}. \tag{4}$$

In order to measure the quality of the approximation, a commonly used cost function is the square of the Euclidean distance between $\boldsymbol{X}$ and $\boldsymbol{A}\boldsymbol{S}$, i.e.,

$$\min_{\boldsymbol{A}\geq 0, \boldsymbol{S}\geq 0} F(\boldsymbol{A}, \boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F^2 \tag{5}$$

where the operator "$\geq$" means that a value will only be greater than or equal to another value.

Among various optimization algorithms, the multiplicative update rule (MUR) was proven by Lee and Seung [28] and widely used because of its low complexity and stable convergence. The loss function in minimization (5) is updated by MUR with the following update rule as

$$\boldsymbol{A} \leftarrow \boldsymbol{A}.*\left(\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}}\right)./\left(\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}\right) \tag{6}$$

$$\boldsymbol{S} \leftarrow \boldsymbol{S}.*\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{X}\right)./\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{S}\right) \tag{7}$$

where the operator "$.*$" and "$./$" represent element-wise multiplication and division, respectively. $(\cdot)^{\mathrm{T}}$ represents matrix transpose.

However, the solution space of NMF model is quite large, and there are many local optimal solutions, because the objective function is not convex with respect to $\boldsymbol{A}$ and $\boldsymbol{S}$ together. This can be confirmed by the obvious truth that the equality $\boldsymbol{A}\boldsymbol{S} = \boldsymbol{A}\boldsymbol{O}\boldsymbol{O}^{-1}\boldsymbol{S}$ holds for any invertible matrix $\boldsymbol{O}$. In order to solve this issue, [12] integrated $L_{1/2}$ sparsity constraint into NMF model, and the $L_{1/2}$-NMF method can obtain sparser and more accurate solutions than other extensions of NMF method. The model of $L_{1/2}$-NMF can be expressed as

$$\min_{\boldsymbol{A}\geq 0, \boldsymbol{S}\geq 0} F(\boldsymbol{A}, \boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F^2 + \mu\|\boldsymbol{S}\|_{1/2} \tag{8}$$

where $\mu \geq 0$ as the penalty of sparse regularization term, which is defined as

$$\|\boldsymbol{S}\|_{1/2} = \sum_{i,j=1}^{M,N} \left(s_{ij}\right)^{1/2} \tag{9}$$

where s_{ij} represents each element in the abundance matrix. According to [12], minimize the loss function in (8), the update rules of endmember matrix $\boldsymbol{A}$ and abundance matrix $\boldsymbol{S}$ are expressed as follows:

$$\boldsymbol{A} \leftarrow \boldsymbol{A}.*\left(\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}}\right)./\left(\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}\right) \tag{10}$$

$$\boldsymbol{S} \leftarrow \boldsymbol{S}.*\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{X}\right)./\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{S} + \frac{\mu}{2}\boldsymbol{S}^{-\frac{1}{2}}\right). \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

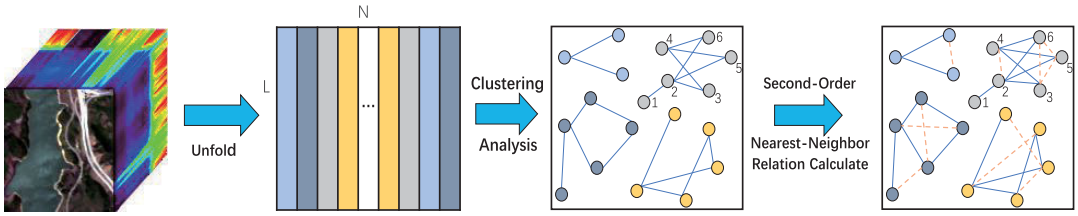


Fig. 1. Schematic of second-order nearest-neighbor relation. Firstly, the hyperspectral data cube is unfolded into a 2-D-matrix, and each column represents a pixel. Then, more spatial information is obtained by finding the second-order adjacent pixel pairs with shared neighborhood structure. (Each circle represents a pixel, and pixels of different colors correspond to different ground-truth. The blue line represents the first-order neighbor connection, and the orange dotted line represents the second-order neighbor connection.)

In formula (11), $\boldsymbol{S}^{-1/2}$ represents a matrix in which each element corresponds to the reciprocal of the square root of each element in matrix $\boldsymbol{S}$.

III. Proposed Model

In this section, we first introduce second-order nearest-neighbor and adaptive feature selection, followed by the overall model. The model shows superiority in both the acquisition of more spatial information and the exploitation of the spectral dimension, which helps to improve the performance of the HU.

A. Second-Order Nearest-Neighbor Relation

According to the principle of local similarity of materials, neighboring pixels tend to share similar abundance. To this end, a large amount of spatial prior information is introduced into the NMF model in order to improve the smoothness of the abundance. The graph regularizer NMF unmixing method constructs an affinity matrix by exploiting the relation between nearest-neighbor pixels, so as to capture the structure of low-dimensional embedded submanifold forms in high-dimensional space to enhance abundance estimation. Specifically, given hyperspectral data $\boldsymbol{X}$, a first-order neighborhood graph can be constructed with pixel points as vertices, and its weight matrix is usually obtained using the following heat kernel function [29]:

$$\boldsymbol{W}_{\mathbf{1}_{ij}} = e^{-\frac{\|x_i - x_j\|^2}{\sigma}} \tag{12}$$

where $\boldsymbol{W}_{\mathbf{1}}$ denote the first-order weight matrix, x_i and x_j denote two pixel points in HSIs, and σ denotes the standard deviation.

The corresponding first-order Laplacian matrix is defined as

$$\boldsymbol{L}_{\mathbf{1}} = \boldsymbol{D}_{\mathbf{1}} - \boldsymbol{W}_{\mathbf{1}} \tag{13}$$

where $\boldsymbol{D}_{\mathbf{1}}$ denotes a diagonal matrix in which $\boldsymbol{D}_{\mathbf{1}_{ii}} = \sum_{j=1}^{N} \boldsymbol{W}_{\mathbf{1}_{ij}}$.

By constructing the above first-order nearest-neighbor graph, an approximate representation of two nearest neighbor pixels in low-dimensional space can be achieved.

However, only considering first-order nearest-neighbor relation in HSIs analysis neglect shared pixel neighborhood structure, leading to missing connections and affecting unmixing performance. In recent years, with the deepening of graph theory research, higher-order proximity relation is introduced into graph learning to process large-scale images and maintain the global spatial structure of images.

To address the shortcomings of first-order nearest-neighbor relation, we introduce second-order nearest-neighbor relation, i.e., pixels with a shared neighborhood structure should be similar. As shown in Fig. 1, the hyperspectral data cube is unfolded into a 2-D hyperspectral matrix. The graph structure of the first-order nearest-neighbor relation is obtained by cluster analysis. If two pixel points are connected by an edge, then these two pixel points have a first-order nearest-neighbor relation and should be closely connected in the embedded low-dimensional space. There is no edge connection between pixel 2 and pixel 4, and so no first-order nearest-neighbor relation exists between these two pixels points, but they have shared neighborhood structure (i.e., 3, 5, 6), and according to the definition of second-order nearest-neighbor relation these two pixel points should also be similar in low-dimensional subspace, that is, existence of second-order nearest-neighbor relation. In graph learning, the similarity between two pixel points is defined in terms of second-order nearest-neighbor relation. In simple terms, the similarity between pixel points is defined by the first-order weight of the shared pixels. If two pixel points are similar to most of the same pixel points, they are similar even though the first-order similarity measure cannot be given.

The weight for second-order nearest-neighbor relation can be expressed as

$$\boldsymbol{W}_{\mathbf{2}_{ij}} = w_{1_i}^{\mathrm{T}} w_{1_j} \tag{14}$$

where w_{1_i} and w_{1_j} denote the ith and jth rows of the first-order weight matrix, respectively. Obviously, when two pixels have many shared neighbors, $\boldsymbol{W}_{\mathbf{2}_{ij}}$ is relatively big. Assuming that there is no pixel connecting x_i and x_j at the same time, the second-order weight matrix between x_i and x_j is zero.

Based on the above analysis, the second-order nearest-neighbor Laplacian matrix is defined as

$$\boldsymbol{L}_{\mathbf{2}} = \boldsymbol{D}_{\mathbf{2}} - \boldsymbol{W}_{\mathbf{2}} \tag{15}$$

where $\boldsymbol{D}_{\mathbf{2}_{ii}} = \sum_{j=1}^{N} \boldsymbol{W}_{\mathbf{2}_{ij}}$. The fused higher-order Laplacian ($\boldsymbol{L}_{\boldsymbol{h}}$), degree matrix ($\boldsymbol{D}_{\boldsymbol{h}}$), and weight matrix ($\boldsymbol{W}_{\boldsymbol{h}}$) are defined as

$$\boldsymbol{L}_{\boldsymbol{h}} = \beta_1 \boldsymbol{L}_{\mathbf{1}} + \beta_2 \boldsymbol{L}_{\mathbf{2}} \tag{16}$$

$$\boldsymbol{D}_{\boldsymbol{h}} = \beta_1 \boldsymbol{D}_{\mathbf{1}} + \beta_2 \boldsymbol{D}_{\mathbf{2}} \tag{17}$$

$$\boldsymbol{W}_{\boldsymbol{h}} = \beta_1 \boldsymbol{W}_{\mathbf{1}} + \beta_2 \boldsymbol{W}_{\mathbf{2}} \tag{18}$$

where β_1 and β_2 are equilibrium parameters.

According to the above analysis, if the two pixel points x_i and x_j are similar, their abundance should also be similar.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

To achieve so, consider minimizing the following function:

$$\begin{aligned}\frac{1}{2}\sum_{i,j=1}^{N}\left\|s_i - s_j\right\|^2 \boldsymbol{W}_{\boldsymbol{h}_{ij}} &= \sum_{i=1}^{N} s_i^{\mathrm{T}} s_i \boldsymbol{D}_{\boldsymbol{h}_{ii}} - \sum_{i,j=1}^{N} s_i^{\mathrm{T}} s_i \boldsymbol{W}_{\boldsymbol{h}_{ij}} \\ &= \mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{D}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right) - \mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{W}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right) \\ &= \mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right). \end{aligned} \tag{19}$$

Therefore, we introduce the higher-order shared neighbor structure into manifold learning, and obtain the higher-order Laplacian graph regularizer.

B. Adaptive Feature Selection

In hyperspectral data, there are large differences in the feature values obtained at different bands. Due to the statistical characteristics of the NMF, the role of tiny features is easily neglected, thus limiting the accuracy of endmember and abundance estimation. Therefore, it is necessary to adequately capture tiny features on each pixel spectrum to guide the NMF-based unmixing process. To this end, we have designed an adaptive weight method that adaptively assigns appropriate weight values to each band during the unmixing process. The wavelength band with high reflectivity is given a smaller weight, while the tiny feature value is given a larger weight. In this way, the contribution of tiny features in data reconstruction can be increased, and the adaptive selection of features can be realized.

The weight matrix is defined as an $L \times L$ nonnegative diagonal matrix $\boldsymbol{\Lambda} \in \mathbb{R}^{L\times L}$, which is defined as follows:

$$\boldsymbol{\Lambda} = \begin{bmatrix} \boldsymbol{\Lambda_{11}} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \boldsymbol{\Lambda_{LL}} \end{bmatrix} \tag{20}$$

where the diagonal term $\boldsymbol{\Lambda_{ll}}$ is the corresponding weight for each band. The initial value of $\boldsymbol{\Lambda}$ is defined as the diagonal matrix of all ones.

The contribution of tiny features is enhanced by adaptive selection of spectral features, balancing the effects of different feature values. By introducing the weight matrix into the model, we obtain the NMF objective function with feature selection

$$\min_{\boldsymbol{A}\geq 0,\boldsymbol{S}\geq 0,\Lambda\geq 0} F(\boldsymbol{A},\boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2. \tag{21}$$

By simple algebraic operations, this can be translated into the following optimized form:

$$\begin{aligned}\min_{\boldsymbol{A}\geq 0,\boldsymbol{S}\geq 0,\boldsymbol{\Lambda}\geq 0} \mathbf{Tr}\left[\boldsymbol{\Lambda}^2\left(\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}} - 2\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}} + \boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\right)\right] \\ = \mathbf{Tr}\left[\boldsymbol{\Lambda}^2\left(\boldsymbol{Z}\boldsymbol{Z}^{\mathrm{T}}\right)\right] \end{aligned} \tag{22}$$

where $\boldsymbol{Z} = \boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}$. The above optimization objective is a constrained quadratic programming (QP) [30], [31] problem of the following form:

$$\begin{aligned}\min_{\boldsymbol{\Lambda}\geq 0} \mathbf{Tr}\left[\boldsymbol{\Lambda}^2\left(\boldsymbol{Z}\boldsymbol{Z}^{\mathrm{T}}\right)\right] &= \sum_{l=1}^{L}\boldsymbol{\Lambda}_l^2\sum_{n=1}^{N} z_{ln}^2 \\ &= \sum_{l=1}^{L}\boldsymbol{\Lambda}_l^2 b_l \end{aligned} \tag{23}$$

where $b_l = \sum_{n=1}^{N} z_{ln}^2$. We achieve adaptive weighting for the spectral space by solving this QP during each iteration.

C. Higher-Order Graph Sparse NMF With Adaptive Feature Selection

The proposed (HGNMF-FS) model can be obtained by integrating higher-order graph regularizer (19) and adaptive feature selection (21) in the NMF model. The specific forms are as follows:

$$\begin{aligned}\min_{\boldsymbol{A}\geq 0,\boldsymbol{S}\geq 0,\boldsymbol{\Lambda}\geq 0} F(\boldsymbol{A},\boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2 + \mu\|\boldsymbol{S}\|_{1/2} \\ + \frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right). \end{aligned} \tag{24}$$

In formula (24), the first term is the fidelity term with feature selection, and the second and third terms are sparse constraint and higher-order Laplacian graph regularization term, respectively. Where μ and λ, respectively, correspond to the regularization term parameters of constraint items, which are utilized to balance the contribution of each item to the model. When the value of a parameter rises, it will improve the effect of the corresponding constraint in the model, and vice versa.

In this article, higher-order Laplacian regularizer and adaptive feature selection, as well as abundance sparsity constraint are introduced, in order to better capture pixel pairs with shared neighborhood, preserve global spatial structure, and enhance the contribution of tiny features in spectral dimension, so as to shrink the solution space and improve the unmixing performance. We utilize the MUR to solve the optimization algorithm of model (24). Next, we will give detailed model optimization steps and algorithm process.

IV. Optimization Algorithm

In this section, we give the process of solution for the optimization algorithm and the problem on implement of the algorithm.

A. Updating Rules

Obviously, the objective function (24) is non-convex for variables $\boldsymbol{A}$ and $\boldsymbol{S}$ together. Therefore, it is not realistic to obtain a global optimal solution. Alternatively, we employ a MUR to obtain local minimal solution by alternately optimizing the objective function. Specifically, we employ this strategy for minimizing the objective function of (24), for each iteration, one matrix is fixed and the other is optimized, thus

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

updating $\boldsymbol{A}$ and $\boldsymbol{S}$ alternately until convergence. Among them, the first $k+1$ iteration was as follows:

$$\begin{cases} \boldsymbol{A}^{k+1} = \underset{\boldsymbol{A}\in\mathbb{R}^{L\times M}}{\arg\min} F\left(\boldsymbol{A}, \boldsymbol{S}^{k}, \boldsymbol{\Lambda}^{k}\right) \\ \boldsymbol{S}^{k+1} = \underset{\boldsymbol{S}\in\mathbb{R}^{M\times N}}{\arg\min} F\left(\boldsymbol{A}^{k+1}, \boldsymbol{S}, \boldsymbol{\Lambda}^{k}\right) \\ \boldsymbol{\Lambda}^{k+1} = \underset{\boldsymbol{\Lambda}\in\mathbb{R}^{L\times L}}{\arg\min} F\left(\boldsymbol{A}^{k+1}, \boldsymbol{S}^{k+1}, \boldsymbol{\Lambda}\right). \end{cases}$$

According to the known constraints, we can get: $\boldsymbol{A}^{L\times M} \geq 0$ and $\boldsymbol{S}^{M\times N} \geq 0$. Let $\boldsymbol{\varphi}^{L\times M} \geq 0$ and $\boldsymbol{\rho}^{M\times N} \geq 0$ be the corresponding Lagrange multipliers, the Lagrange function Γ corresponding to F in (24) can be expressed as

$$\begin{aligned} \Gamma &= \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2 + \mu\|\boldsymbol{S}\|_{\frac{1}{2}} + \frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right) \\ &\quad + \mathbf{Tr}\left(\boldsymbol{\varphi}\boldsymbol{A}^{\mathrm{T}}\right) + \mathbf{Tr}\left(\boldsymbol{\rho}\boldsymbol{S}^{\mathrm{T}}\right) \\ &= \frac{1}{2}\mathbf{Tr}\left(\boldsymbol{\Lambda}\boldsymbol{X}\boldsymbol{X}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\right) - \mathbf{Tr}\left(\boldsymbol{\Lambda}\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\right) \\ &\quad + \frac{1}{2}\mathbf{Tr}\left(\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\right) + \mu\|\boldsymbol{S}\|_{\frac{1}{2}} \\ &\quad + \frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}}\boldsymbol{S}^{\mathrm{T}}\right) + \mathbf{Tr}\left(\boldsymbol{\varphi}\boldsymbol{A}^{\mathrm{T}}\right) + \mathbf{Tr}\left(\boldsymbol{\rho}\boldsymbol{S}^{\mathrm{T}}\right). \end{aligned} \tag{25}$$

Here, the second equation utilizes the definition of Frobenius norm $\|\boldsymbol{Q}\|_F = (\mathbf{Tr}(\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}))^{1/2}$ and the properties of two traces $\mathbf{Tr}(\boldsymbol{A}\boldsymbol{B}) = \mathbf{Tr}(\boldsymbol{B}\boldsymbol{A})$, $\mathbf{Tr}(\boldsymbol{A}) = \mathbf{Tr}(\boldsymbol{A}^{\mathrm{T}})$. The subproblems of $\boldsymbol{A}$, $\boldsymbol{S}$, and $\boldsymbol{\Lambda}$ are solved as follows.

1) Optimization of $\boldsymbol{A}$: The partial derivative equation (25) to $\boldsymbol{A}$ can be given by

$$\frac{\partial\Gamma}{\partial\boldsymbol{A}} = -\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}} + \boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}} + \boldsymbol{\varphi}. \tag{26}$$

In [32], we have learned about Karush–Kuhn–Tucker (KKT) conditions

$$\boldsymbol{\varphi}_{\boldsymbol{lm}}\boldsymbol{A}_{\boldsymbol{lm}} = 0. \tag{27}$$

Applying it to (26), we can obtain

$$\left(-\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}} + \boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}\right).*\boldsymbol{A} = 0. \tag{28}$$

Therefore, the update rules for the endmember matrix $\boldsymbol{A}$ can be obtained by

$$\boldsymbol{A} \leftarrow \boldsymbol{A}.*\left(\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X}\boldsymbol{S}^{\mathrm{T}}\right)./\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S}\boldsymbol{S}^{\mathrm{T}}. \tag{29}$$

2) Optimization of $\boldsymbol{S}$: Its partial derivative with respect to $\boldsymbol{S}$ can be calculated by

$$\frac{\partial\Gamma}{\partial\boldsymbol{S}} = -\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X} + \boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S} + \frac{\mu}{2}\boldsymbol{S}^{-\frac{1}{2}} + \lambda\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}} + \boldsymbol{\rho}. \tag{30}$$

According to the KKT conditions, we can obtain as

$$\boldsymbol{\rho}_{\boldsymbol{mn}}\boldsymbol{S}_{\boldsymbol{mn}} = 0. \tag{31}$$

Applying it to (30), the following equation is obtained:

$$\left(-\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X} + \boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S} + \frac{\mu}{2}\boldsymbol{S}^{-\frac{1}{2}} + \lambda\boldsymbol{S}\boldsymbol{L}_{\boldsymbol{h}}\right).*\boldsymbol{S} = 0. \tag{32}$$

By using transformations and separations while using $\boldsymbol{L}_{\boldsymbol{h}} = \boldsymbol{D}_{\boldsymbol{h}} + \boldsymbol{W}_{\boldsymbol{h}}$, the $\boldsymbol{S}$ update rule can be obtained by

$$\boldsymbol{S} \leftarrow \boldsymbol{S}.*\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{X} + \lambda\boldsymbol{S}\boldsymbol{W}_{\boldsymbol{h}}\right)./\left(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda}\boldsymbol{A}\boldsymbol{S} + \frac{\mu}{2}\boldsymbol{S}^{-\frac{1}{2}} + \lambda\boldsymbol{S}\boldsymbol{D}_{\boldsymbol{h}}\right). \tag{33}$$

3) Optimization of $\boldsymbol{\Lambda}$: We fix $\boldsymbol{A}$ and $\boldsymbol{S}$. The optimization problem for $\boldsymbol{\Lambda}$ can be written as

$$\min_{\boldsymbol{\Lambda}\geq 0} \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2. \tag{34}$$

Clearly, this is a standard QP problem. We achieve adaptive weighting of the feature space by solving this QP problem during each iteration.

Finally, the detailed optimization algorithm to solve the proposed HGNMF-FS method is summarized in Algorithm 1.

Algorithm 1 HGNMF-FS Algorithm

Input: The observed data $\boldsymbol{X} \in \mathbb{R}^{L\times N}$ and the number of endmembers M; The parameters μ, λ, β_1, β_2 and MaxIter;

Output: Endmember matrix $\boldsymbol{A} \in \mathbb{R}^{L\times M}$ and abundance matrix $\boldsymbol{S} \in \mathbb{R}^{M\times N}$;

Initialization: Initialize $\boldsymbol{A}$ and $\boldsymbol{S}$ by VCA and FCLS, respectively; Construct the weight matrix $\boldsymbol{\Lambda}$ with all one;

repeat:
  Calculate the endmember matrix $\boldsymbol{A}$ by using Eq. (29);
  Update abundance matrix $\boldsymbol{S}$ by solving problem (33);
  Update weight matrix $\boldsymbol{\Lambda}$ by solving QP problem (34);
  Update iteration: k ← k + 1.
until some stopping criterion is satisfied.
return results

B. Implementation Issues

To implement effectively the proposed unmixing method, several issues need to be considered.

The first issue concerns the initialization of the $\boldsymbol{A}$ and $\boldsymbol{S}$, since an inappropriate initialization may cause the algorithm to be stuck in local minima. We use an unsupervised endmember extraction method, vertex component analysis (VCA) to initialize $\boldsymbol{A}$, and employ fully constrained least squares (FCLS) to obtain the initial $\boldsymbol{S}$.

The second issue is how to ensure that the two LMM constraints are satisfied in the unmixing process. According to the MUR, we conclude that the nonnegative constraint can be achieved as long as the initial $\boldsymbol{A}$ and $\boldsymbol{S}$ are nonnegative. In each iteration, we replace $\boldsymbol{X}$ and $\boldsymbol{A}$ with their augmented forms to ensure that ASC. Their augmented forms are written as

$$\overline{\boldsymbol{X}} = \begin{pmatrix} \boldsymbol{X} \\ \delta\mathbf{1}_N^{\mathrm{T}} \end{pmatrix}, \quad \overline{\boldsymbol{A}} = \begin{pmatrix} \boldsymbol{A} \\ \delta\mathbf{1}_M^{\mathrm{T}} \end{pmatrix} \tag{35}$$

where δ is a parameter to control the weight of the ASC in the objective function. According to [33], a larger δ results in a more precise ASC, but also leads to a slower convergence process. Therefore, we set $\delta = 15$ to balance these opposite effects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

The third issue comes from the $L_{1/2}$ sparse operator, specifically the term $\boldsymbol{S}^{-1/2}$ in (33). To prevent the denominator from being zero, we use a predefined threshold $\xi = 10^{-4}$ replace the entries in $\boldsymbol{X}$ and $\boldsymbol{A}$ when whose values are smaller than ξ.

In addition, we set two stop conditions. The first one is a value of error tolerance. If the error does not exceed the set tolerance value for ten consecutive iterations, the iteration ends. We set $\epsilon = 1\text{e}^{-4}$. The second one is the maximum number of iterations. In this model, our maximum number of iterations is set to 3000, which is fully guaranteed to converge.

One final issue about how to determine the number of endmember. The question of how to determine the number of endmembers is a separate topic in HU. We estimate the number of endmember either by manual setup or by a validated method, such as virtual dimensionality [34] and HySime [35].

C. Computational Complexity Analysis

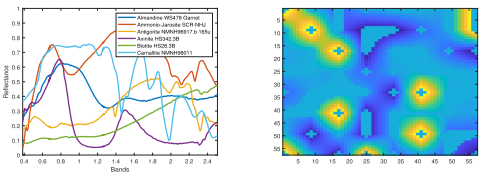
Here, we briefly analyze the computational complexity of the proposed HGNMF-FS method. In order to make the analysis clearer, the time of floating-point calculation has been counted for each iteration.

As shown in Algorithm 1, the most costly steps of the proposed method are to solve $\boldsymbol{A}$ and $\boldsymbol{S}$ problem (i.e., steps 2 and 3). From (29) and (33) corresponding to the MUR of HGNMF-FS and (6) and (7) representing the MUR of the standard NMF, it is clear that the main difference between HGNMF-FS and the standard NMF method is the weight matrix $\boldsymbol{\Lambda}$. Therefore, we analyzed the computational cost of the proposed algorithm on the basis of the standard NMF results. First, the computational complexity of the standard NMF is $O(LNP)$. In the update process, we need to note that different matrix multiplication orders are also important [22], such as $(\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda})\boldsymbol{X}$ and $\boldsymbol{\Lambda}^{\mathrm{T}}(\boldsymbol{\Lambda}\boldsymbol{X})$, leading to different computational costs in different orders. The former needs $O(L^3 + L^2N)$, whereas the latter costs $O(L^2N)$. This means that $(\boldsymbol{\Lambda}^{\mathrm{T}}\boldsymbol{\Lambda})\boldsymbol{X}$ is much more complicated than $\boldsymbol{\Lambda}^{\mathrm{T}}(\boldsymbol{\Lambda}\boldsymbol{X})$. In addition, each row of the first-order weight matrix $\boldsymbol{W_1}$ has only k non-zero elements. The floating-point addition and multiplication of the second-order weight matrix $\boldsymbol{W_2}$ actually takes k^2N times, and the calculation of $\boldsymbol{SW_h}$ actually takes $K^2N + 2kPN$ times. The same is true of $\boldsymbol{SD_h}$. Additionally, the cost of computing $\boldsymbol{S}^{-(1/2)}$ and $\boldsymbol{\Lambda}$ is $O(NP^2)$ and $O(L)$, respectively.

Note that the cost of constructing the K-nearest neighbor graph is $O(N^2L)$. Considering all factors together, if the iteration ends at step u, the total cost is $O(u(2L^2N + LNP + (NP)^2 + L) + N^2L + k^2N + 2kPN)$.

V. Experimental Results

In this section, the experiments on synthetic and real dataset are conducted to verify the effectiveness of the proposed method. We compare the VCA[1]-FCLS [4], [33], $L_{1/2}$-NMF [12], GLNMF [19], R-CONMF[2] [36], TV-RSNMF[3] [14], MV-NTF-TV[4] [37], SC-NTF [38], SUnSAL-TV[5] [7] and CUSUnSAL [39] with the methods proposed in this article. These comparison algorithms are effective and advanced. In addition, Table I shows the optimal parameter settings of the comparison algorithm. All the experiments are carried out on a Windows 10 system with 3.2-GHz AMD Ryzen 7 5800H CPU and 16-GB RAM using MATLAB2021a.



(a) (b)

Fig. 2. Synthetic hyperspectral data (DC1). (a) Endmember spectrum. (b) Simulated image.

The results were evaluated using the spectral angle distance (SAD) and root mean square error (RMSE) [40]. They are defined as follows:

$$\mathrm{SAD} = \sum_{i=1}^{M} \arccos\left(\frac{\boldsymbol{e}_i^{\mathrm{T}}\widetilde{\boldsymbol{e}}_i}{\|\boldsymbol{e}_i\| \cdot \|\widetilde{\boldsymbol{e}}_i\|}\right) \tag{36}$$

$$\mathrm{RMSE} = \left(\frac{1}{N}\sum_{i=1}^{N}\|\boldsymbol{a}_i - \widetilde{\boldsymbol{a}}_i\|_2^2\right)^{1/2} \tag{37}$$

where $\boldsymbol{e}_i$ and $\widetilde{\boldsymbol{e}}_i$ mean the ith estimate endmember vector and corresponding truth endmember, respectively; $\boldsymbol{a}_i$ and $\widetilde{\boldsymbol{a}}_i$ denote the ith estimated abundance vector and the ground-truth abundance related to it, respectively.

A. Experiment on Synthetic Dataset

Data Generation: The generation of synthetic data includes two steps: endmember selection and abundance estimation. To demonstrate the unmixing performance of HGNMF-FS, we select the following two synthetic datasets.

DC1: For the simulated data cube 1 (DC1), we randomly selected six spectra from the U.S. Geological Survey (USGS)[6] digital spectral library [41], as shown in Fig. 2, for the following experiments. As shown in Fig. 3, the synthetic data were generated using the method provided in [13]. The process of generated synthetic data is then divided into five steps: 1) the image with a size of 64 × 64 is divided into 8 × 8 patches; 2) each patch is populated with the same endmember randomly selected in the endmember sets; 3) the mixed pixel is generated with 8 × 8 low-pass filter; 4) we set parameter $a = 0.8$ to limit the purity of the mixed pixels; and 5) Gaussian white noise is added to the synthesized data, and the signal-to-noise ratio (SNR) is set to 20 dB.

DC2: For the simulated data cube 2 (DC2), we first randomly selected five spectra from the USGS digital spectral library, then generate simulated abundance maps according to the method provided in [8], and finally synthesize the endmember and abundances into simulated hyperspectral data according to the LMM, as shown in Fig. 4. This simulated image has 75 × 75 pixels with 188 bands were generated

[1]http://www.lx.it.pt/~bioucas/code.htm
[2]http://www.lx.it.pt/~jun/
[3]https://prowdiy.github.io/weihe.github.io/
[4]http://www.xiongfuli.com/cv/cn.html
[5]https://www2.umbc.edu/rssipl/people/aplaza/
[6]http://speclab.cr.usgs.gov/spectral.lib06

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

TABLE I
PARAMETER SETTING OF COMPARE ALGORITHMS

Algorithm	VCA-FCLS	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	R-CONMF	SUnSAL-TV	CLSUnSAL
Parameter	None	$\lambda = 0.1$ $\mu = 0.1$	$\lambda = 0.35$	$\lambda = 0.01$ $\tau = 0.01$	$\delta = 15$ $\lambda = 1$	$\mu = 10$ $\lambda = 0.05$	$\alpha = 10^{-8}$ $\beta = 10^{-1}$	$\lambda = 1e-3$ $\lambda_{TV} = 3e-3$	$\lambda = 3e-3$

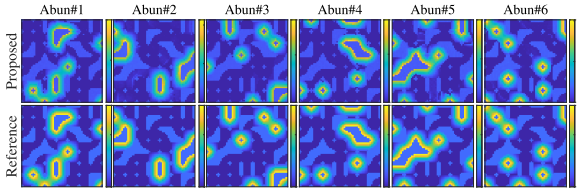
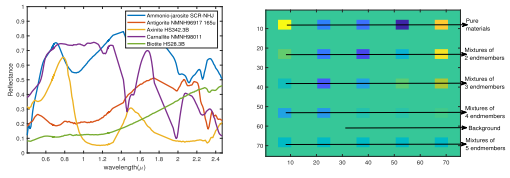
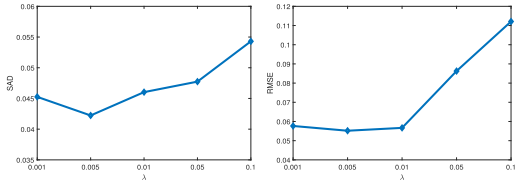


Fig. 3. References and estimated abundance maps on DC1.



(a) (b)

Fig. 4. Synthetic hyperspectral data (DC2). (a) Endmember spectrum. (b) Simulated image.



(a) (b)

Fig. 5. Performance analysis of the proposed method with respect to parameters λ in terms of (a) SAD and (b) RMSE.

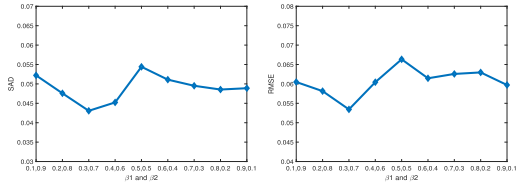
using five spectral signatures according to LMM, and applied ASC to the pixels. There are not only pure pixel regions in the image, but also mixed regions constructed using mixtures ranging two, three, four, and five endmembers.

All methods have the experiment that were repeated twenty times to ensure a reliable comparison. In addition, all the parameters of the compared algorithms are consistent with references.

1) Experiments on DC1:

a) Experiment 1 (Parameter analysis): In this experiment, three critical parameters λ, β_1 and β_2 in formula (24) are taken into account when SNR = 20 dB. First, we analyze the parameter λ while fixing $\mu = 1e^{-4}$ values. As shown in Fig. 5, when λ changes from 0.001 to 0.1, the SAD and RMSE values are relatively stable in this interval. When λ is above 0.005, both SAD and RMSE values rise clearly. Therefore, we choose the optimal parameter $\lambda = 0.005$ for subsequent experiments.

Next, we consider the problem of the proportion of first-order nearest-neighbors and second-order nearest-neighbors in the higher-order Laplacian graph regularizer. In Fig. 6, the effect of different proportion on the experimental results is given. We can clearly observe that the highest SAD and RMSE values are reached at values of $\beta_1 = 0.5$ and $\beta_2 = 0.5$, respectively. With the increasing proportion of second-order



(a) (b)

Fig. 6. Performance analysis of the proposed method with respect to first-order and second-order nearest-neighbors occupancy in terms of (a) SAD and (b) RMSE.

nearest-neighbors or first-order nearest-neighbors, SAD and RMSE values both decrease, indicating that both first-order and second-order nearest-neighbors can play a role in HU. In comparison, the lowest SAD and RMSE values are reached when the second-order neighborhood information is more dominant, which means that the second-order nearest-neighbor relation has a large credit in exploring spatial nearest-neighbor information. Therefore, the optimal values of SAD and RMSE for β_1 and β_2 were at 0.3 and 0.7, respectively. Based on the above analysis, we set $\beta_1 = 0.3$ and $\beta_2 = 0.7$ in the rest of the experiments.

b) Experiment 2 (Robustness analysis to noise): The main goal of this experiment is to evaluate the robustness of the proposed method at different noise levels, i.e., SNR = 15, 20, 25, 30, and 35 dB. It is worth noting that all methods in the experiment use the same initialization conditions and set the same number of endmembers. Tables II and III represent the SAD and RMSE values for the all compare methods with different SNR conditions, respectively. In order to make the results clearer, we use bold to mark the best unmixing results. Obviously, all methods exhibit the similar trend, with the increase of SNR, the SAD and RMSE values obtained by all algorithms is declining, which indicates that the performance of all methods is improving and also the importance of denoising in the HU process. Interestingly, regardless of the noise intensity, our proposed method obtains the lowest SAD and RMSE values compared to other methods, showing better stability and robustness. For example, when SNR = 20 dB, the value of SAD obtained by our HGNMF-FS method is lower about 0.012 and 0.037 than those obtained by GLNMF and $L_{1/2}$-NMF method, respectively. Thus, further proof of our idea that second-order nearest-neighbor relation can effectively complement the shared neighbor structure where first-order nearest-neighbor relation is missing. In this way, the convergence of homogeneous regions is enhanced and the solution space is reduced more efficiently.

c) Experiment 3 (Image size of synthetic data): The purpose of this experiment is tantamount to investigate the sensitivity of the proposed method to different image sizes. The sizes of synthetic data used in this experiment are 25 × 25, 36 × 36, 49 × 49, 64 × 64, 81 × 81, and 100 × 100,

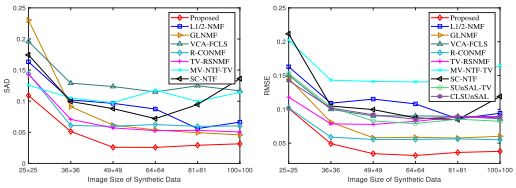
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULTS OF SAD WITH DIFFERENT NOISE LEVELS ON DC1

SNR	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
SNR=15	0.1379	0.0948	0.0579	0.0788	0.0755	0.0831	0.1026	**0.0511**
SNR=20	0.1385	0.0594	0.0559	0.0809	0.0646	0.0747	0.0949	**0.0437**
SNR=25	0.1275	0.0551	0.0539	0.0799	0.0560	0.0719	0.0894	**0.0422**
SNR=30	0.1279	0.0468	0.0482	0.0769	0.0539	0.0681	0.0876	**0.0414**
SNR=35	0.1262	0.0449	0.0464	0.0690	0.0518	0.0672	0.0872	**0.0411**

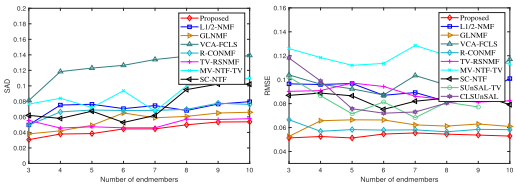
TABLE III
RESULTS OF RMSE WITH DIFFERENT NOISE LEVELS ON DC1

SNR	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	SUnSAL-TV	CLSUnSAL	Proposed
SNR=15	0.0984	0.0707	0.0704	0.0995	0.1012	0.1967	0.0943	0.0866	0.0933	**0.0685**
SNR=20	0.0899	0.0546	0.0605	0.0894	0.0928	0.1966	0.0905	0.0862	0.0837	**0.0545**
SNR=25	0.0949	0.0623	0.0633	0.0982	0.0910	0.1954	0.0895	0.0819	0.0840	**0.0594**
SNR=30	0.0899	0.0573	0.0603	0.0893	0.0900	0.1952	0.0889	0.0776	0.0799	**0.0532**
SNR=35	0.0874	0.0564	0.0574	0.0875	0.0859	0.1901	0.0878	0.0719	0.0772	**0.0474**



(a) (b)

Fig. 7. Comparing the algorithms of simulation data at different size levels according to (a) SAD and (b) RMSE.

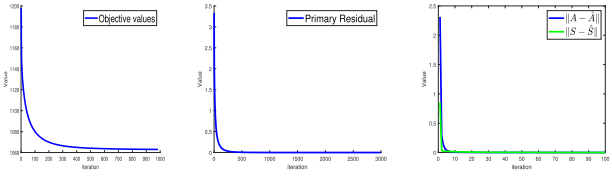


(a) (b)

Fig. 8. Comparing the algorithms of simulation data at different numbers of endmembers according to (a) SAD and (b) RMSE.

respectively. The number of endmembers in synthetic data is 4, and the other parameters are set as SNR = 20 dB. As shown in Fig. 7, the proposed method has the best performance in composite images of different sizes. We can clearly that the unmixing performance of most algorithms can be divided into two stages as the number of pixels increases. In the first stage, from 25 × 25 to 49 × 49, the performance of the algorithms is improved, because it is difficult for the algorithm based on NMF to obtain enough statistical information when the image is too small. The second stage is from 64 × 64 to 100 × 100 when the performance of all methods has reached a steady state. Meanwhile, with the increase of the image size, the performance of our proposed method is gradually improved, mainly because the second-order neighbor relation is used to supplement the pixel pairs with the missing connection of the first-order neighbor relation, and more shared neighborhood spatial structure information will be mined to assist in HU.

d) Experiment 4 (Different numbers of endmembers): In this experiment, we evaluate the performance of all methods when the data consists of different numbers of endmembers. The SNR was set to 20 dB in the synthetic data and the number of endmembers was set to 3–10. In Fig. 8, the performance of all methods tends to a decreasing situation as the number of endmembers increases. In general, our proposed method outperforms other methods for all different numbers of endmembers. Interestingly, as the number of endmembers continues to increase, the larger the difference value between the SAD and RMSE values of the proposed algorithm and the algorithm GLNMF, which can reflect the obvious advantages



(a) (b) (c)

Fig. 9. Convergence curves. (a) Objective values. (b) Primary residual. (c) Estimate the residuals of $\hat{\boldsymbol{A}}$ and $\hat{\boldsymbol{S}}$.

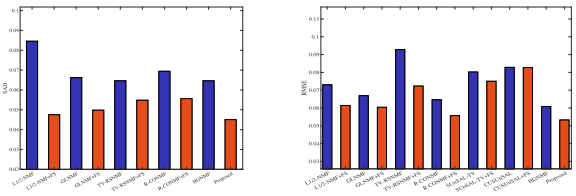
of the proposed method. This implies that our proposed HGNMF-FS can handle more complex HSIs.

e) Experiment 5 (Convergence analysis): This experiment summarizes the convergence analysis of the proposed algorithm. The convergence of the algorithm in this article can be analyzed and proved in a similar way to [15]. We set the number of endmember to 6 and SNR = 20 dB. Fig. 9 plots the curve of the objective function value and the primary residual term. It can be seen that the objective function value decreases rapidly after the first few iterations and converges to a stable value after about 100 iterations. The primary residual term also shows a decreasing trend, which is very well in line with our expected results.

Meanwhile, we also give an iterative residual curve of estimated for endmembers and abundance. As shown in Fig. 9(c), we can see that after 100 iterations, the residuals of the two matrices approach to zero.

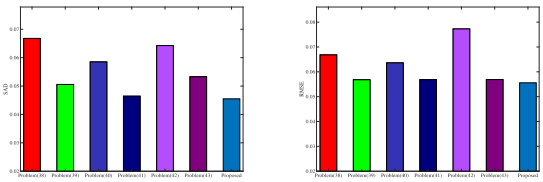
f) Experiment 6 (Feature selection analyse): This experiment is to verify the effectiveness of the feature selection method. Here, the number of endmembers is set to 6 and SNR = 20 dB. As shown in Fig. 10, we analyze the influence

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 10. Comparison of performance of (a) SAD and (b) RMSE after adding feature selection to different algorithms.



(a) (b)

Fig. 11. Comparing (a) SAD and (b) RMSE of different ablation experiments.

of feature selection on the performance of the algorithm. It is clear that the SAD and RMSE values decreased to varying degrees for all methods when the feature selection method was added. In addition, the importance of the influence of different spectral band feature values on the results in HU is verified. The impact of different feature values is balanced by introducing adaptive weight to adaptively select spectral features to enhance the contribution of tiny features. The model with feature selection shows superior performance after adding all kinds of priors, which indicates that our model with adaptive feature selection has better generalization ability.

g) Experiment 7 (Ablation analysis): Here, we design three ablation experiments to discuss the higher-order nearest-neighbor relation and adaptive feature selection. First, we discuss the different unmixing effects by adding adaptive feature selection and not adding adaptive feature selection on the base algorithm NMF. Corresponding two loss functions are as follows:

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2 \tag{38}$$

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S}\|_F^2. \tag{39}$$

Then, we once again discuss the different unmixing effects of only first-order nearest-neighbor relation and only second-order nearest-neighbor relation with adaptive feature selection. The corresponding two loss functions are as follows:

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2+\mu\|\boldsymbol{S}\|_{\frac{1}{2}}+\frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_1\boldsymbol{S}^{\mathrm{T}}\right) \tag{40}$$

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{\Lambda}(\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S})\|_F^2+\mu\|\boldsymbol{S}\|_{\frac{1}{2}}+\frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_2\boldsymbol{S}^{\mathrm{T}}\right). \tag{41}$$

Finally, we discuss the performance of adding first-order nearest-neighbor relation and second-order nearest-neighbor relation without adaptive feature selection. Its corresponding cost function is expressed as follows:

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S}\|_F^2+\mu\|\boldsymbol{S}\|_{\frac{1}{2}}+\frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_1\boldsymbol{S}^{\mathrm{T}}\right) \tag{42}$$

$$\min_{\boldsymbol{A}\geq\boldsymbol{0},\boldsymbol{S}\geq\boldsymbol{0}} \frac{1}{2}\|\boldsymbol{X}-\boldsymbol{A}\boldsymbol{S}\|_F^2+\mu\|\boldsymbol{S}\|_{\frac{1}{2}}+\frac{\lambda}{2}\mathbf{Tr}\left(\boldsymbol{S}\boldsymbol{L}_2\boldsymbol{S}^{\mathrm{T}}\right). \tag{43}$$

We test them for the synthetic data with $M = 6$ and SNR $= 20$ dB. In Fig. 11, the ablation experiment of regularizers is performed by ignoring the higher-order nearest-neighbor relation and adaptive feature selection in our proposed model. Obviously, in (38) and (39), the performance has been greatly improved by adding adaptive feature selection method to the basic algorithm NMF. Problems (40) and (41) only consider the first-order or second-order nearest-neighbor when there is adaptive feature selection. Both methods have certain advantages after adding adaptive feature selection to the first-order or second-order nearest-neighbor relation. However, (42) and (43) study the first-order and second-order nearest-neighbor relation without adaptive feature selection, and the performance of these two methods is not particularly ideal without adaptive feature selection. But when we only consider the second-order nearest-neighbor relation, we still get better performance than only considering the first-order nearest-neighbor relation. Finally, we can draw a conclusion that the adaptive feature selection and high-order nearest-neighbor relation in our model have a positive impact on the unmixing performance.

2) Experiments on DC2: We now study the performance of HGNMF-FS and the compared algorithms on DC2. Tables IV and V show the SAD and RMSE values of different algorithms in simulated data (DC2) when the SNR ratios are 10, 20, 30, and 40 dB, respectively. We can clearly see that with the increase of SNR, the SAD and RMSE values obtained by all methods show a downward trend. In the high SNR environment, the geometric method (VCA) also achieves good unmixing performance due to the existence of pure pixels. However, when SNR $= 10$ dB, the performance of VCA-FCLS method is degraded. Through the fusion of second-order neighborhood relations and the addition of adaptive feature selection, we have achieved good performance under different SNRs. In Fig. 12, we give the abundance diagram of five endmembers with different algorithms under the condition of SNR $= 20$ dB. It can be clearly seen from the figure that our method can always give the ground object distribution estimation correctly, and the obtained abundance map is smooth and clear, which is highly consistent with the real abundance. Under the condition of high SNR, the abundance estimated by the proposed method is highly consistent with the real abundance.

The average execution time of each algorithm is given in Table VI to evaluate its time efficiency. The VCA-FCLS method has the fastest running speed because it is a non-iterative algorithm. Compared with other NMF methods, our method has less time consumption, and can further smooth the abundance and improve the unmixing accuracy.

B. Real Hyperspectral Data

In this section, we will use real datasets to verify the effectiveness of the method proposed in this article. We will

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
RESULTS OF SAD WITH DIFFERENT NOISE LEVELS ON DC2

SNR	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
SNR=10	0.0742	0.0744	0.0835	0.0833	0.0932	0.3258	0.0733	**0.0721**
SNR=20	0.0206	0.0196	0.0660	0.0656	0.0510	0.2362	0.0199	**0.0195**
SNR=30	0.0068	0.0069	0.0624	0.0640	0.0456	0.2925	0.0067	**0.0062**
SNR=40	0.0021	0.0030	0.0619	0.0615	0.0448	0.2915	0.0033	**0.0019**

TABLE V
RESULTS OF RMSE WITH DIFFERENT NOISE LEVELS ON DC2

SNR	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	SUnSAL-TV	CLSUnSAL	Proposed
SNR=10	0.0575	0.0507	0.0707	0.0727	0.0772	0.0934	0.0582	0.0619	0.0510	**0.0476**
SNR=20	0.0285	0.0191	0.0513	0.0512	0.0343	0.0933	0.0197	0.0186	0.0188	**0.0185**
SNR=30	0.0069	0.0069	0.0440	0.0436	0.0259	0.0889	0.0073	0.0067	0.0065	**0.0064**
SNR=40	0.0025	0.0029	0.0436	0.0432	0.0253	0.0633	0.0031	**0.0021**	0.0027	**0.0021**

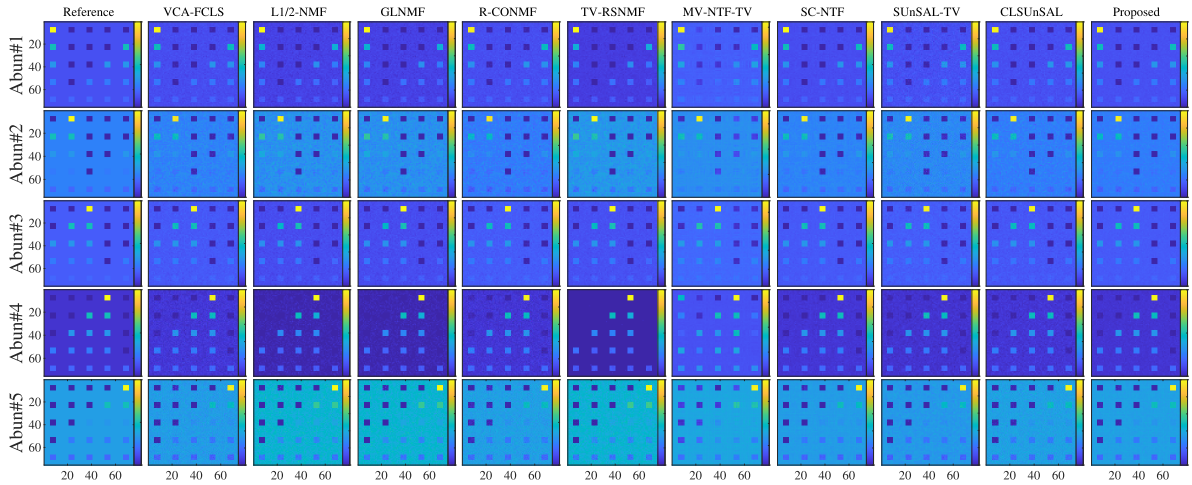


Fig. 12. References and estimated abundance maps by different unmixing algorithms on DC2 for endmembers 1–5.

TABLE VI
AVERAGESSING TIME (IN SECONDS) OF EACH COMPETING METHOD ON SIMULATED DATASETS (DC2)

Methods	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	SUnSAL-TV	CLSUnSAL	Proposed
Time[s]	0.3930	0.6560	34.2410	28.7470	22.5310	133.8990	112.4460	3.485163	0.156859	0.6170

verify it on four real datasets, namely Samson,[7] Jasper Ridge,[8] Urban,[9] and AVIRIS Cuprite.[10] All methods use the same initialization method. The experiment is repeated twenty times to ensure a reliable comparison.

1) Samson Dataset: Samson is a dataset widely used for HU. It contains 95 × 95 pixels, and each pixel has 156 bands in the range of [0.041 0.889] μm. According to [42] and [43], this region contains three materials, i.e., tree, water, and soil.

The Samson dataset is shown in Fig. 13 by HGNMF-FS estimation of the endmember signatures the reference signatures. Clearly, the signature estimation of the three materials is consistent with the reference signature, which shows the effectiveness of the proposed method. Due to the unavailability of the true abundances of the real dataset [10], Fig. 14 only shows the abundance maps estimated by HGNMF-FS.

[7]http://www.escience.cn/people/feiyunZHU/Dataset_GT.html
[8]:http://www.escience.cn/system/file?fileId=68574
[9]http://www.agc.army.mil/
[10]http://aviris.jpl.nasa.gov/data/free_data.html

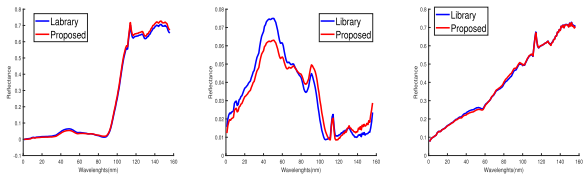


Fig. 13. Comparison of the reference spectra with the endmember signatures extracted by the proposed method on the Samson dataset. Followed by tree, water, soil.

Table VII shows the SAD values obtained by our proposed method, as well as the comparisons for each material. As shown in table, the proposed method obtained better performance in both soil and trees. Most importantly, the overall performance of the newly proposed algorithm is the best, which demonstrates the superiority of our proposed method over other methods.

2) Jasper Ridge Dataset: In the field of HU, this dataset is widely used in research. Owing to the fact that Jasper Ridge data images are too complex. In this experiment, we research

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

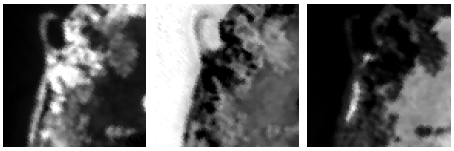


Fig. 14. Abundance maps of the different endmembers obtained using HGNMF-FS on the Samson dataset. (Left to right) Tree, water, soil.

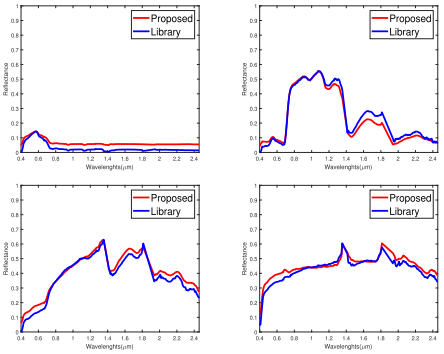


Fig. 15. Comparison of the USGS library spectra with the endmember signatures extracted by HGNMF-FS on the Jasper Ridge dataset. (Left to right and top to bottom) Tree, soil, water, and road.

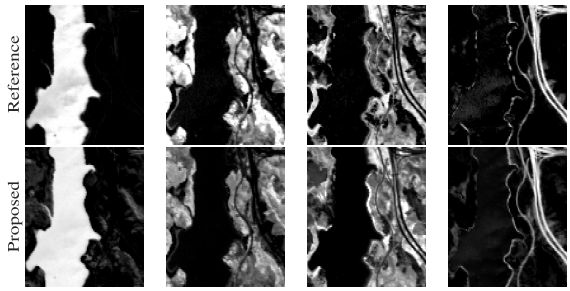


Fig. 16. Abundance maps of the different endmembers obtained using HGNMF-FS on the Jasper Ridge dataset. (Left to right) Water, tree, soil, and road.

in a subspace of 100 × 100 pixels with each pixel remaining 198 bands after removal of noise bands. In [44], four references endmembers are considered in the sub-scene: tree, water, soil, and road.

The endmember signature extracted using the proposed method and the reference endmember signature obtained from the USGS library is given in Fig. 15. It can be clearly seen that the endmembers extracted by the method are all close to the reference endmembers, indicating that the method is effective. Table VIII shows the SAD values obtained by all compare methods. From the table, we can note that most of the materials extracted using the method in this article have the lowest SAD values. Fig. 16 shows the grayscale abundance maps obtained by our proposed method, where the brighter pixels represent the higher abundance of the corresponding endmembers, and it can be seen that the estimated endmembers always agree with the real endmembers, verifying the effectiveness of the proposed method.

3) HYDICE Urban Dataset: Urban is one of the most widely used hyperspectral data in HU research. There are 307 × 307 pixels, and each pixel contains 210 bands, which cover the wavelength range of 0.4–2.4 μm [45], [46]. After the channels 1–4, 76, 87, 101–111, 136–153, and 198–210 were removed (due to intensive water vapor and atmospheric effects), 162 channels remained. According to previous

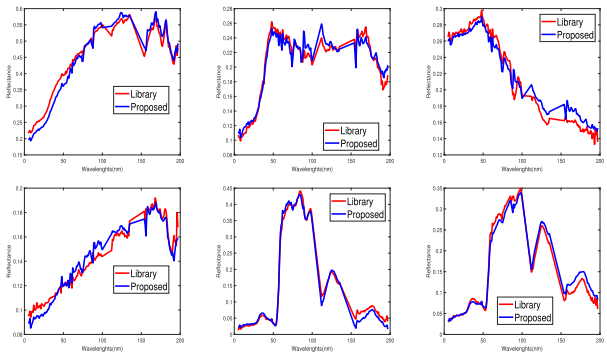


Fig. 17. Comparison of the USGS library spectra with the endmember signatures extracted by HGNMF-FS on the urban dataset. (Left to right and top to bottom) Asphalt-road, grass, tree, Roof#1, Roof#2, and concrete-road.

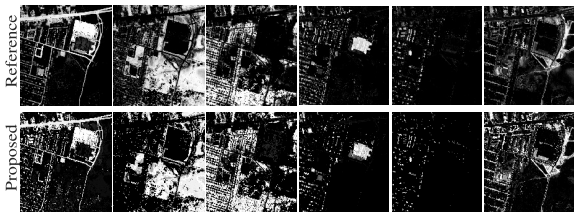


Fig. 18. Abundance maps of different endmembers obtained by HGNMF-FS method on hyperspectral dataset urban. (Left to right) Asphalt-road, grass, tree, Roof#1, Roof#2, and concrete-road.

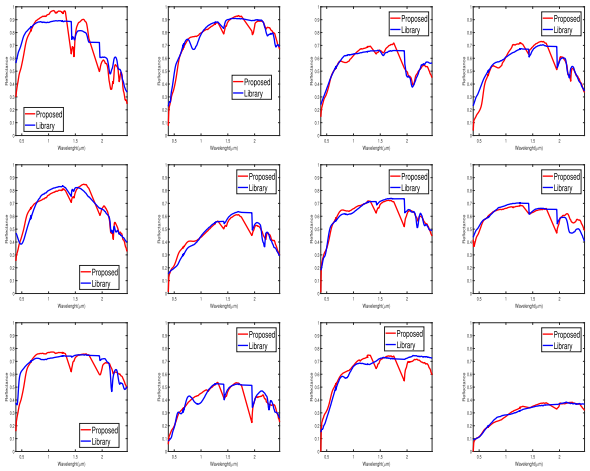


Fig. 19. Comparison of the USGS library spectra with the endmember signatures extracted by HGNMF-FS on the AVIRIS Cuprite dataset. (Left to right and top to bottom) Alunite, andradite, buddingtonite, chalcedony, dumortierite, Kaolinite#1, Kaolinite#2, montmorillonite, muscovite, nontronite, pyrope, and sphene.

research works, this scene consists of six materials, namely asphalt-road, roof#1, roof#2, grass, tree, and concrete-road.

The results are summarized in Table IX, Figs. 17 and 18. As can be seen from Table IX, most of the endmember signatures extracted using the method in this article have the highest accuracy and the lowest SAD mean value. Fig. 17 shows the estimated endmembers and their references. It can be clearly seen that the results of the proposed method are very close to the referenced endmembers. To large-sized HSIs, our proposed method obtains favorable results in all six endmembers, which show the superiority and accuracy of the proposed method for processing large-sized and complex HSIs. In Fig. 18, we can clearly observe that the abundance is consistent with the distribution of ground-truth.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
SAD RESULTS ON THE SAMSON DATA

Algorithms	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
Tree	0.1531	0.0498	0.0141	0.0266	0.0240	0.0141	0.0289	**0.0134**
Soil	0.0487	0.0813	0.0186	0.0292	0.0244	0.0148	0.0442	**0.0144**
Water	**0.1296**	0.3568	0.2496	0.2767	0.2860	0.1604	0.1656	0.1470
Mean	0.1104	0.1626	0.0941	0.1108	0.1115	0.0631	0.0795	**0.0583**

TABLE VIII
SAD RESULTS ON THE JASPER RIDGE DATA

Algorithms	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
Tree	0.1016	**0.0407**	0.0578	0.0373	0.0553	0.0814	0.0578	0.0603
Soil	0.4844	0.4715	0.4889	0.5404	0.5312	0.5190	0.4642	**0.4597**
Water	0.1902	0.1416	0.0669	0.0610	0.0948	0.2247	0.0846	**0.0593**
Road	0.2492	0.1871	0.1836	0.1757	0.2368	0.1741	0.1984	**0.0829**
Mean	0.2564	0.1871	0.1836	0.1757	0.2295	0.2498	0.2012	**0.1656**

TABLE IX
SAD RESULTS ON THE HYDICE URBAN DATA

Algorithms	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
Asphalt-road	0.2418	0.1697	0.1422	0.1633	0.0490	0.0912	0.0472	**0.0468**
Grass	0.2104	0.1721	0.0859	0.0924	0.0869	0.1006	0.0938	**0.0793**
Tree	0.4011	0.1897	0.1433	0.0920	0.0839	0.1009	0.0993	**0.0619**
Roof#1	0.2606	0.2303	0.0488	0.0494	0.0953	0.1216	0.1041	**0.0471**
Roof#2	0.5150	0.2748	0.0491	0.0578	0.1613	0.1267	0.1551	**0.0430**
Concrete-road	0.2495	0.5796	0.0565	0.0558	0.1983	0.1563	0.2096	**0.0468**
Mean	0.3131	0.2694	0.0876	0.0853	0.1124	0.1162	0.1184	**0.0542**

TABLE X
SAD RESULTS ON THE AVIRIS CUPRITE DATA

Algorithms	VCA-FCLS	R-CONMF	GLNMF	$L_{1/2}$-NMF	TV-RSNMF	MV-NTF-TV	SC-NTF	Proposed
Alunite	0.1224	0.1610	0.1555	0.1225	0.1494	0.1249	0.1113	**0.1110**
Andradite	0.0891	0.3281	**0.0669**	0.0701	0.0983	0.0843	0.1341	0.0768
Buddingtonite	0.1123	0.3282	0.0947	0.1116	0.1014	0.0952	0.0965	**0.0914**
Dumortierite	0.1229	0.1414	0.1755	0.1316	0.1537	0.1226	0.1213	**0.1098**
Kaolinite#1	0.0911	0.1126	**0.0627**	0.0732	0.0791	0.0833	0.0895	0.0699
Kaolinite#2	0.0969	0.1209	0.1198	0.0999	0.0959	0.0998	0.1265	**0.0932**
Montmorillonite	0.0602	0.1272	0.0568	**0.0541**	0.0843	0.0729	0.1232	0.0638
Nontronite	0.0912	0.2007	**0.0878**	0.0993	0.1064	0.1015	0.1034	0.0923
Pyrope	0.1809	0.1083	**0.0685**	0.1104	0.1119	0.1668	0.0990	0.1589
Sphene	0.1201	0.2367	0.2504	0.2212	0.1647	0.0907	0.0879	**0.0598**
Chalcedony	0.1210	0.2492	0.0893	0.0921	0.0833	0.1301	0.0882	**0.0831**
Muscovite	0.1510	0.1808	0.1351	0.1596	0.08865	0.1276	**0.1204**	0.1771
Mean	0.1130	0.1920	0.1136	0.1096	0.1097	0.1083	0.1084	**0.0989**

4) AVIRIS Cuprite Dataset: The fourth experiment of real hyperspectral data is conducted on AVIRIS Cuprite data. This real dataset is a benchmark dataset, which is used to study the performance of unmixing algorithm on large and complex datasets [47]. Finally, we selected a cube with the size of $250 \times 191 \times 188$ as experimental data to ensure the effectiveness of all algorithms. As assumed by most experiments based on real data, we reduced the number of endmembers to 12 by ignoring the slight differences between similar minerals. According to the latest separation results [48], 12 types of minerals are: "#1 Alunite," "#2 Andradite," "#3 Buddingtonite," "#4 Chalcedony," "#5 Dumortierite," "#6 Kaolin#1," "#7 Kaolin#2," "#8 Montmorillonite," "#9 Muscovite," "#10 Nontronite," "#11 Pyrope," and "#12 Sphene."

In Fig. 19, we compare the reference endmembers selected from the USGS library with those estimated by HGNMF-FS on the AVIRIS Cuprite dataset. Similarly, the 12 minerals obtained by our proposed method are in a high degree of consistency with the reference endmember spectral curves. In addition, the abundance maps estimated by HGNMF-FS are shown in Fig. 20, where the brighter pixels represent the higher abundance of the endmembers. As shown in Table X, the mean SAD values corresponding to each mineral for different algorithms. We can see that the best performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

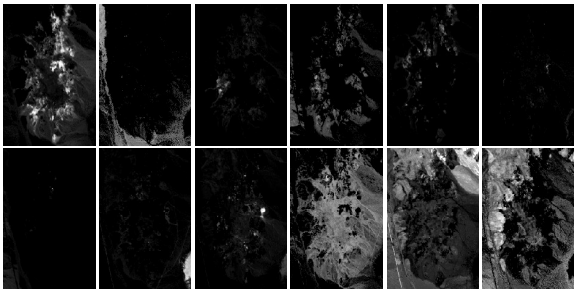


Fig. 20. Abundance maps of the different endmembers obtained using HGNMF-FS on the AVIRIS Cuprite dataset. (Left to right and top to bottom) Alunite, andradite, buddingtonite, chalcedony, dumortierite, Kaolinite#1, Kaolinite#2, montmorillonite, muscovite, nontronite, pyrope, and sphene.

can be obtained with HGNMF-FS in most minerals. Most importantly, the average SAD value of the method in this article is the smallest, which confirms the superiority of the method in the case of multiple endmembers.

VI. Conclusion

In this article, we proposed a higher-order graph regularizer NMF with adaptive feature selection HU method. The method introduces the second-order nearest-neighbor relation in graph learning to complement the pixel points in the first-order neighborhood with missing shared neighborhood structure information and to enhance the aggregation of homogeneous regions. Then the Laplacian graph regularizer is constructed by combining the first-order and second-order nearest-neighbor relation information, which is used to better maintain the global spatial structure. In addition, adaptive weight is introduced in the data reconstruction term to enhance the contribution of tiny features by adaptively selecting spectral features to balance the influence of different feature values and further improve the performance of unmixing. Experimental results on synthetic and real datasets demonstrated the effectiveness of the proposed method and achieved better HU performance than other methods.

In future work, we will use second-order nearest neighbor information on a tensor-based model to explore more spatial information in the 3-D cube. In addition, we will improve the performance of the algorithm by introducing a fast method instead of time-consuming operations.

Acknowledgment

The authors would like to thank the editors and the two anonymous reviewers for their constructive, detailed, and helpful comments and suggestions regarding this article. They would also like to thank the authors of the VCA-FCLS, R-CONMF, TV-RSNMF, MV-NTF-TV, and SUnSAL-TV algorithms for generously sharing their code and data.

References

[1] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[2] J. Li et al., "Multiple feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1592–1606, Mar. 2015.

[3] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[4] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[5] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[6] J. Li, A. Agathos, D. Zaharie, J. M. Bioucas-Dias, A. Plaza, and X. Li, "Minimum volume simplex analysis: A fast algorithm for linear hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 9, pp. 5067–5082, Sep. 2015.

[7] M.-D. Iordache, J. Bioucas-Dias, and A. Plaza, "Sparse unmixing of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2014–2039, Jun. 2011.

[8] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Total variation spatial regularization for sparse hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4484–4502, Nov. 2012.

[9] H.-C. Li, X.-R. Feng, D.-H. Zhai, Q. Du, and A. Plaza, "Self-supervised robust deep matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513214.

[10] X. Feng, H. Li, J. Li, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using sparsity-constrained deep nonnegative matrix factorization with total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6245–6257, Oct. 2018.

[11] D. Donoho and V. Stodden, "When does non-negative matrix factorization give a correct decomposition into parts?" in *Proc. Adv. Neural Inf. Process. Syst. Conf. (NIPS)*, 2004, pp. 1–8.

[12] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

[13] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[14] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[15] K. Qu and W. Bao, "Multiple-priors ensemble constrained nonnegative matrix factorization for spectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 963–975, 2020.

[16] J. B. Tenenbaum, V. D. Silva, and J. C. Langford, "A global geometric framework for nonlinear dimensionality reduction," *Science*, vol. 290, no. 5500, pp. 2319–2323, Dec. 2000.

[17] S. T. Roweis and L. K. Saul, "Nonlinear dimensionality reduction by locally linear embedding," *Science*, vol. 290, no. 5500, pp. 2323–2326, Dec. 2000.

[18] Y. Duan, H. Huang, and Y. Tang, "Local constraint-based sparse manifold hypergraph learning for dimensionality reduction of hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 613–628, Jan. 2021.

[19] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[20] B. Rathnayake, E. M. M. B. Ekanayake, K. Weerakoon, G. M. R. I. Godaliyadda, M. P. B. Ekanayake, and H. M. V. R. Herath, "Graph-based blind hyperspectral unmixing via nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 9, pp. 6391–6409, Sep. 2020.

[21] X. Lu, L. Dong, and Y. Yuan, "Subspace clustering constrained sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3007–3019, May 2020.

[22] F. Zhu, Y. Wang, S. Xiang, B. Fan, and C. Pan, "Structured sparse method for hyperspectral unmixing," *ISPRS J. Photogramm. Remote Sens.*, vol. 88, pp. 101–118, Feb. 2014.

[23] J. Tang, M. Qu, M. Wang, M. Zhang, J. Yan, and Q. Mei, "LINE: Large-scale information network embedding," in *Proc. 24th Int. Conf. World Wide Web*, May 2015, pp. 1067–1077.

[24] X. Liu, P. Song, C. Sheng, and W. Zhang, "Robust multi-view nonnegative matrix factorization for clustering," *Digit. Signal Process.*, vol. 123, Apr. 2022, Art. no. 103447.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.

[25] J.-Y. Wang, I. Almasri, and X. Gao, "Adaptive graph regularized nonnegative matrix factorization via feature selection," in *Proc. Int. Conf. Pattern Recognit.*, 2012, pp. 963–966.

[26] X. Lu, H. Wu, and Y. Yuan, "Double constrained NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2746–2758, May 2014.

[27] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[28] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, Oct. 1999.

[29] M. Belkin and P. Niyogi, "Laplacian eigenmaps and spectral techniques for embedding and clustering," in *Advances in Neural Information Processing System*, vol. 14. Cambridge, MA, USA: MIT Press, 2002, pp. 585–591.

[30] D. R. Hush, P. Kelly, C. Scovel, and I. Steinwart, "Qp algorithms with guaranteed accuracy and run time for support vector machines," *J. Mach. Learn. Res.*, vol. 7, pp. 733–769, May 2006.

[31] M. J. Best and J. Hlouskova, "An algorithm for portfolio optimization with transaction costs," *Manage. Sci.*, vol. 51, no. 11, pp. 1676–1688, Nov. 2005.

[32] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.

[33] D. C. Heinz and C.-I. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[34] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[35] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[36] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.

[37] F. Xiong, Y. Qian, J. Zhou, and Y. Y. Tang, "Hyperspectral unmixing via total variation regularized nonnegative tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2341–2357, Apr. 2019.

[38] H. Li, S. Liu, X. Feng, and S. Zhang, "Sparsity-constrained coupled non-negative matrix–tensor factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5061–5073, Aug. 2020.

[39] M.-D. Iordache, J. M. Bioucas-Dias, and A. Plaza, "Collaborative sparse regression for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 341–354, Jan. 2014.

[40] W. Wang, Y. Qian, and Y. Y. Tang, "Hypergraph-regularized sparse NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 681–694, Feb. 2016.

[41] R. Clark, G. Swayze, A. Gallagher, T. King, and W. Calvin, "The U.S. geological survey digital spectral library: Version 1: 0.2 to 3.0 microns," U.S. Geol. Surv., Reston, VA, USA, Open File Rep. 93-592, 1993.

[42] Y. E. Salehani and S. Gazor, "Smooth and sparse regularization for NMF hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 8, pp. 3677–3692, Aug. 2017.

[43] F. Zhu, Y. Wang, B. Fan, S. Xiang, G. Meng, and C. Pan, "Spectral unmixing via data-guided sparsity," *IEEE Trans. Image Process.*, vol. 23, no. 12, pp. 5412–5427, Dec. 2014.

[44] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

[45] X. Liu, W. Xia, B. Wang, and L. Zhang, "An approach based on constrained nonnegative matrix factorization to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 757–772, Feb. 2011.

[46] S. Jia and Y. Qian, "Constrained nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 161–173, Jan. 2009.

[47] Y. Yuan, M. Fu, and X. Lu, "Substance dependence constrained sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 2975–2986, Jun. 2015.

[48] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Blind hyperspectral unmixing using total variation and ℓ_q sparse regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 11, pp. 6371–6384, Nov. 2016.



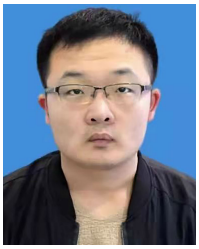
Kewen Qu (Member, IEEE) received the B.Eng. degree in computer science and technology from Henan University, Kaifeng, China, in 2008, the M.Sc. degree in computer science from North Minzu University, Yinchuan, China, in 2012, and the Ph.D. degree from the School of Computer and Information, Hefei University of Technology, Hefei, China, in 2020.

His research interests include hyperspectral images (HSIs) processing, machine learning, and pattern recognition.



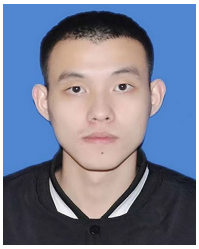
Zhenqing Li (Student Member, IEEE) received the B.Eng. degree in computer science and technology from the Henan University of Engineering, Zhengzhou, China, in 2021. He is currently pursuing the M.S. degree with the School of Computer Science and Engineering, North Minzu University, Yinchuan, China.

His research interests include hyperspectral unmixing and machine learning.



Chenyang Wang received the B.Eng. degree in computer science and technology from the North University of China, Taiyuan, China, in 2019. He is currently pursuing the M.S. degree with the School of Computer Science and Engineering, North Minzu University, Yinchuan, China.

His research interests include hyperspectral classification and neural networks.



Fangzhou Luo received the B.Eng. degree in computer science and technology from the Chongqing Institute of Engineering, Chongqing, China, in 2022. He is currently pursuing the M.S. degree with the School of Computer Science and Engineering, North Minzu University, Yinchuan, China.

His research interests include hyperspectral unmixing.



Wenxing Bao (Member, IEEE) received the B.Eng. degree in industrial automation from Xidian University, Xi'an, China, in 1993, and the M.Sc. degree in electrical engineering and the Ph.D. degree in electronic science and technology from Xi'an Jiaotong University, Xi'an, in 2001 and 2006, respectively.

He is currently a Professor with North Minzu University, Yinchuan, China. His research interests include digital image processing, and remote sensing image classification and fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 20:31:19 UTC from IEEE Xplore. Restrictions apply.